

Stage	Layer	Configuration	Output
Input x4	Morgan / RDKit / MACCS / ChemBERTa	2048 / 100 / 167 / 768	per-branch
Encoders x4	Dense+BatchNorm+Dropout	128, GELU, drop 0.25	4 x 128
Fusion	Concatenate+BatchNorm	-	512
To-sequence	Dense -> Reshape	512 -> (64x8)	(64, 8)
Conv x2	residual Conv1D	32 filters, k=3, GELU	(64, 32)
Attention	squeeze-excite	-	(64, 32)
Pool	MaxPooling1D	pool 2	(32, 32)
Recurrent	Bidirectional LSTM	64 units (->128)	128
Head	res-Dense->Drop->Dense	64, GELU	64
Output	Dense	1, sigmoid	P(active)